

Pre-screening

Participant	009
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	Yes	No
Do you or have you had a history of migraine headaches?	☐	☒
Do you or have you had a history of claustrophobia?	☐	☒
Do you or have you had a history of frequent or severe motion sickness?	☐	☒
Do you or have you had a history of any health problems (e.g., seizures, diabetes, heart problems, vertigo) that affect your ability to drive?	☐	☒
If you are a female, are you or is there a possibility that you might be pregnant?	☐	☒

Demographics

1. What is your race/ethnicity?	☒ White / Caucasian
2. What is your gender?	☒ Male
3. How old are you?	34
4. What is the highest level of education you have completed?	☒ High school graduate of GED
5. Which option best describes your current home?	☒ It is owned or being bought by you or someone in household.
6. How do you pay for healthcare?	☒ Self-pay, out of pocket
8. Which best corresponds to your current work situation?	☒ Not working or looking for work.
9. What is your combined family income for the past 12 months (from all sources)	☒ \$100,000- \$149,000

Timeline Followback

Visit 1

Visit 1 Date

02-10-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 2

Visit 2 Date

05-22-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 3

Visit 3 Date

06-02-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 4

Visit 4 Date

07-08-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 5

Visit 5 Date

07-16-2025

Have you used opioids or any other substances during the assessment period?

☒ Yes

Phone

Participant # _____

009

System Usability Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently	☐	☒	☐	☐	☐
2. I found the system unnecessarily complex	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☒	☐
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☒
8. I found the system very cumbersome to use.	☒	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☒	☐
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this app experience.	☒	☐	☐	☐	☐
2. The time I spent in this app just slipped away.	☒	☐	☐	☐	☐
3. I was absorbed in the app task.	☐	☒	☐	☐	☐
4. I felt frustrated while using the app.	☒	☐	☐	☐	☐

5. I found this app to be confusing.	☒	☐	☐	☐	☐
6. Using this app was taxing.	☒	☐	☐	☐	☐
7. The app is attractive.	☐	☐	☐	☒	☐
8. The app is aesthetically appealing.	☐	☐	☐	☐	☒
9. The app appealed to my senses.	☐	☐	☐	☒	☐
10. Using this app was worthwhile.	☐	☐	☐	☒	☐
11. The therapy experience was rewarding.	☐	☐	☒	☐	☐
12. I felt interested in the therapy tasks.	☐	☐	☒	☐	☐

iPAL Application and Intervention Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☒	☐	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☒	☐	☐
3. The videos were videos easy to understand.	☐	☐	☒	☐	☐
4. The breathing exercises were smooth and easy to do.	☐	☐	☒	☐	☐

AR

Participant ID _____

009

- | | |
|---|--|
| 1. Which option describes your current home? | ☒ It is owned or being bought by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Self-pay, out of pocket |
| 3. Which best corresponds to your current work situation? | ☒ Working full time. |

iGROUP Presence Questionnaire

- | | |
|--|--|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Somewhat Disagree |
| 2. I felt that the virtual world surround me. | ☒ Completely Disagree |
| 3. I felt like I was perceiving pictures. | ☒ Somewhat Agree |
| 4. I did not feel present in the virtual space. | ☒ Completely Agree |
| 5. I had a sense of acting in the virtual space, rather than operating something form the outside. | ☒ Somewhat Disagree |
| 6. I felt present in the virtual space | ☒ Somewhat Disagree |
| 7. How aware were you of the real world surrounding you while navigating in the virtual world? | ☒ Somewhat unaware |
| 8. I was not aware of my real environment. | ☒ Somewhat Agree |
| 9. I still paid attention to my real environment. | ☒ Somewhat Agree |
| 10. I was completely captivated by the virtual world. | ☒ Neither Agree nor Disagree |
| 11 How real did the virtual world seem to you? | ☒ Somewhat real |
| 12. How much did your experience in the virtual environment seem consistent with your real-world experience? | ☒ Somewhat Disagree |
| 13. How real did the virtual world seem to you? | ☒ Somewhat real |
| 14. The virtual world seemed more realistic than the real world. | ☒ Somewhat Disagree |

System Usability Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☐	☐	☒	☐
2. I found the system unnecessarily complex.	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☐	☒	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☒	☐
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☒	☐
8. I found the system very cumbersome to use.	☒	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☒	☐
10. I needed to learn a lot of things before I could get going with this system.	☐	☒	☐	☐	☐

User Engagement Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this augmented reality experience	☐	☒	☐	☐	☐
2. The time I spent in augmented reality just slipped away.	☐	☒	☐	☐	☐
3. I was absorbed in the therapy task.	☐	☐	☐	☒	☐
4. I felt frustrated while using the headset.	☒	☐	☐	☐	☐
5. I found this system to be confusing.	☐	☒	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☐	☒	☐

8. The augmented reality is aesthetically appealing.	☐	☐	☐	☒	☐
9. The augmented environment appealed to my senses.	☐	☐	☐	☒	☐
10. Using this system was worthwhile.	☐	☐	☒	☐	☐
11. The therapy experience was rewarding.	☐	☐	☐	☒	☐
12. I felt interested in the therapy tasks.	☐	☐	☐	☒	☐

iPAL Application and Intervention Evaluation AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☒	☐	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☒	☐
3. The videos were clear and easy to understand.	☐	☐	☐	☐	☒

VR

Participant ID _____ 009

- | | |
|---|--|
| 1. Which option describes your current home? | ☒ It is owned or being bought by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Self-pay, out of pocket |
| 3. Which best corresponds to your current work situation? | ☒ Working full time. |

iGroup Presence Scale

- | | |
|--|---|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Somewhat Agree |
| 2. I felt that the virtual world surrounded me. | ☒ Somewhat Agree |
| 3. I felt like I was perceiving pictures. | ☒ Completely Agree |
| 4. I did not feel present in the virtual space. | ☒ Completely Disagree |
| 5. I had a sense of acting in the virtual space, rather than operating something from the outside. | ☒ Completely Agree |
| 6. I felt present in the virtual space. | ☒ Completely Agree |
| 7. How aware were you of the real world surrounding you while navigating in the virtual world? | ☒ Completely unaware |
| 8. I was not aware of my real environment. | ☒ Completely Agree |
| 9. I still paid attention to my real environment. | ☒ Somewhat Agree |
| 10. I was completely captivated by the virtual world. | ☒ Completely Agree |
| 11. How real did the virtual world seem to you? | ☒ Somewhat real |
| 12. How much did your experience in the virtual environment seem consistent with your real-world experience? | ☒ Somewhat Agree |
| 13. How real did the virtual world seem to you? | ☒ Very real |
| 14. The virtual world seemed more realistic than the real world. | ☒ Somewhat Disagree |

System Usability Scale VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☐	☐	☐	☒
2. I found the system unnecessarily complex.	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☐	☒	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☒	☐
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☒
8. I found the system very cumbersome to use.	☒	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☒	☐
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this virtual experience	☐	☐	☐	☐	☒
2. The time I spent in the virtual world just slipped away.	☐	☐	☐	☒	☐
3. I was absorbed in the therapy task.	☐	☐	☐	☐	☒
4. I felt frustrated while using the headset.	☒	☐	☐	☐	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☐	☒	☐

8. The virtual environment is aesthetically appealing.	☐	☐	☐	☐	☒
9. The virtual environment appealed to my senses.	☐	☐	☐	☐	☒
10. Using this system was worthwhile.	☐	☐	☐	☐	☒
11. The therapy experience was rewarding.	☐	☐	☐	☐	☒
12. I felt interested in the therapy tasks.	☐	☐	☐	☐	☒

iPAL Application and Intervention Evaluation VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☒	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☒	☐
3. The videos were clear and easy to understand.	☐	☐	☐	☐	☒

MR

Participant ID _____

009

- | | |
|---|--|
| 1. Which option describes your current home? | ☒ It is owned or being bought by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Self-pay, out of pocket |
| 3. Which best corresponds to your current work situation? | ☒ Working full time. |

iGROUP Presence Questionnaire MR

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|--|--|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Somewhat Agree |
| 2. I felt that the virtual world surrounded me. | ☒ Somewhat Agree |
| 3. I felt like I was perceiving pictures. | ☒ Somewhat Agree |
| 4. I did not feel present in the virtual space. | ☒ Neither Agree nor Disagree |
| 5. I had a sense of acting in the virtual space, rather than operating something from the outside. | ☒ Somewhat Agree |
| 6. I felt present in the virtual space. | ☒ Somewhat Agree |
| 7. How aware were you of the real world surround you while navigating in the virtual world? | ☒ Very aware |
| 8. I was not aware of my real environment. | ☒ Completely disagree |
| 9. I still paid attention to my real environment. | ☒ Completely Agree |
| 10. I was completely captivated by the virtual world. | ☒ Somewhat Agree |
| 11. How real did the virtual world seem to you? | ☒ Somewhat real |
| 12. How much your experience in the virtual environment seem consistent with your real-world experience? | ☒ Neither Agree nor Disagree |
| 13. How real did the virtual world seem to you? | ☒ Somewhat real |
| 14. The virtual world seemed more realistic than the real world. | ☒ Somewhat disagree |

System Usability Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☐	☐	☒	☐
2. I found the system unnecessarily complex.	☐	☒	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☒	☐
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☒	☐
6. I thought there was too much inconsistency in this system.	☐	☒	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☒	☐
8. I found the system very cumbersome to use.	☒	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☒	☐
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this mixed-reality experience.	☐	☐	☐	☒	☐
2. The time I spent in mixed reality just slipped away.	☐	☐	☐	☒	☐
3. I was absorbed in the therapy task.	☐	☐	☐	☒	☐
4. I felt frustrated while using the headset.	☒	☐	☐	☐	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☐	☒	☐

8. The mixed-reality is aesthetically appealing.	☐	☐	☐	☒	☐
9. The blended environment appealed to my senses.	☐	☐	☐	☒	☐
10. Using this system was worthwhile.	☐	☐	☐	☒	☐
11. The therapy experience was rewarding.	☐	☐	☐	☒	☐
12. I felt interested in the therapy tasks.	☐	☐	☐	☒	☐

iPAL Application and Intervention Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☒	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☒	☐	☐
3. The videos were clear and easy to understand.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☐	☒

Exit Interview

1. Which option describes your current home?	☒ It is owned or being bought by you or someone in household
2. How do you pay for healthcare?	☒ Self-pay, out of pocket
3. Which best corresponds to your current work situation?	☒ Working full time.
1. You experienced Cognitive Behavioral Therapy and Heart Rate Variance interventions in virtual, augmented, and mixed realities. How would you describe your experiences with each of those headsets?	believes that both augmented and mixed realities were cool and had structure, but enjoyed the virtual reality the most. believes that it would have the most impact on recovery treatment and was the most immersive.
2. Do you think these types of technologies have a place in OUD treatment and management? If YES, how? If NO, why not?	believes that virtual reality would have the most positive impact. believes that it would allow participants to have a space to relax and listening to the videos while focusing.
3. How would you describe the iPAL application, in terms of a cravings management tool?	believes that the app covers the bases of the aspects that are important for recovery, but believes that in order for it to be effective, the patient must be fully aware of their needs.
4. How would you describe the interventions (CBT, HRV)? Were the interventions appropriate for the intensity level and triggers you reported?	CBT - believes that it was helpful. notes that it was hard to initially settle in and listen to what the individual in the video was saying, but believes that everything that she said was important and helpful. HRV - very helpful. believes that the the exercise allowed him to get focused and calm and allowed him to settle into the type of reality that he was experiencing. believes that it should be the first intervention that is done, prior to watching CBT videos.
5. Do you think the iPAL app, or similar digital interventions delivered via phone should be integrated into OUD treatment plans? If YES, how and in what capacity or role? if NO, why not?	yes, but only if a patient is motivated and willing to share their accurate emotions honestly.
6. If iPAL or a similar app was available, would you download the app and incorporate it into your OUD treatment and management plan?	yes, would definitely give it a try, especially after experiencing it through the study. could envision self utilizing the app at least once a week.